



Object Detection and Semantic Segmentation of Aerial and Satellite Imagery with Deep Learning Models

*Data Science and Machine Learning
Postgraduate Thesis Defense Presentation*

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Presentation Outline

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Motivation & Problem Statement

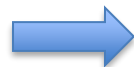
Detection and segmentation answer different operational questions.

Detection

- ✓ What is present and roughly where.
- ✓ Fast to annotate and run.
- ✓ Box geometry cannot describe precise object shape.

Segmentation

- ✓ Where every object pixel lies.
- ✓ Expensive to annotate.
- ✓ Required for footprints, area estimation, vector layers, and geospatial analytics.



Real geo-applications need both

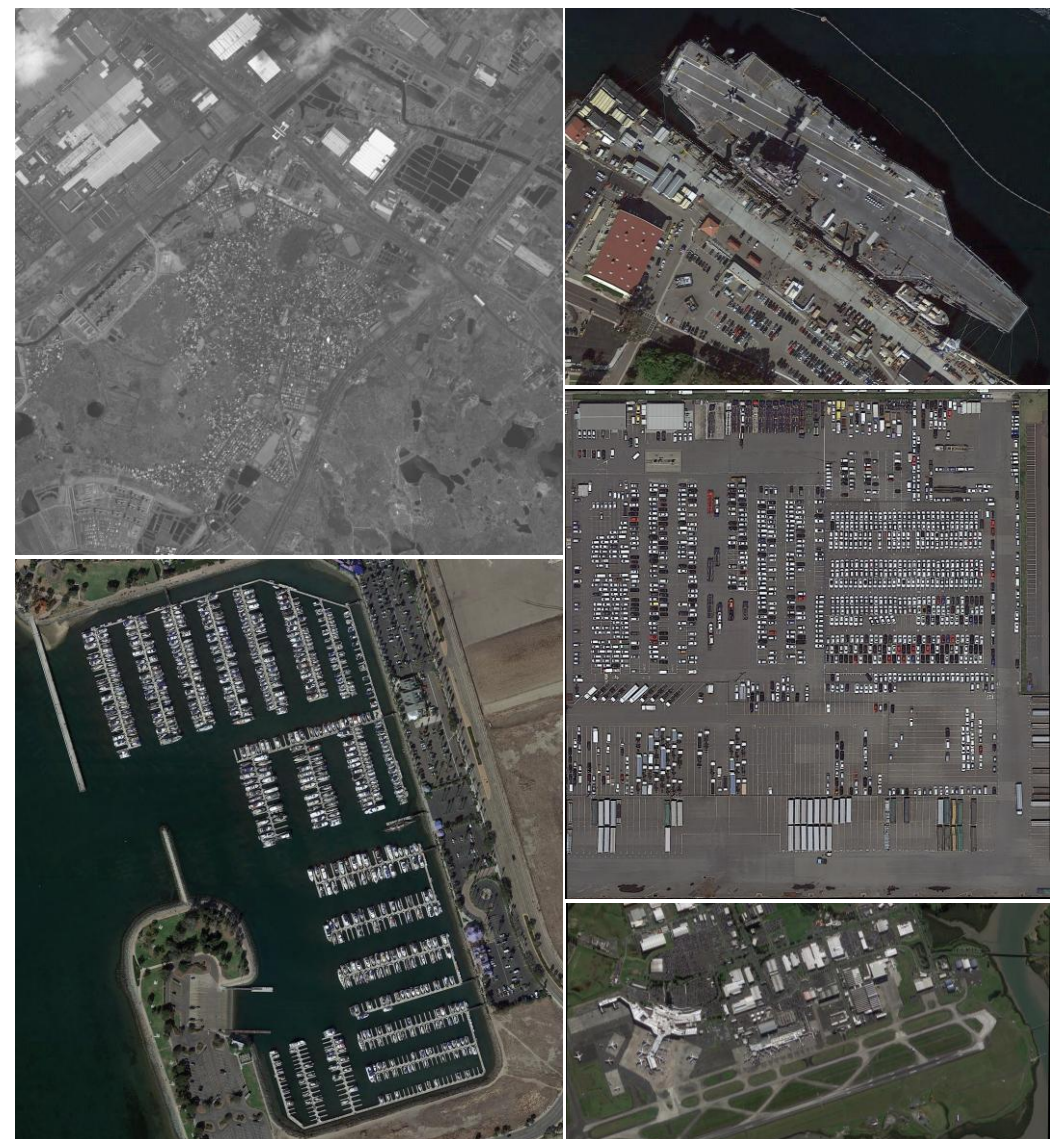
- ✓ Disaster response & damage assessment
- ✓ Infrastructure monitoring (e.g. roads, buildings)
- ✓ Maritime awareness & vessel tracking
- ✓ Precision agriculture & land-use mapping
- ✓ Urban planning & dynamic change detection



Motivation & Problem Statement

Challenges in Aerial Imagery

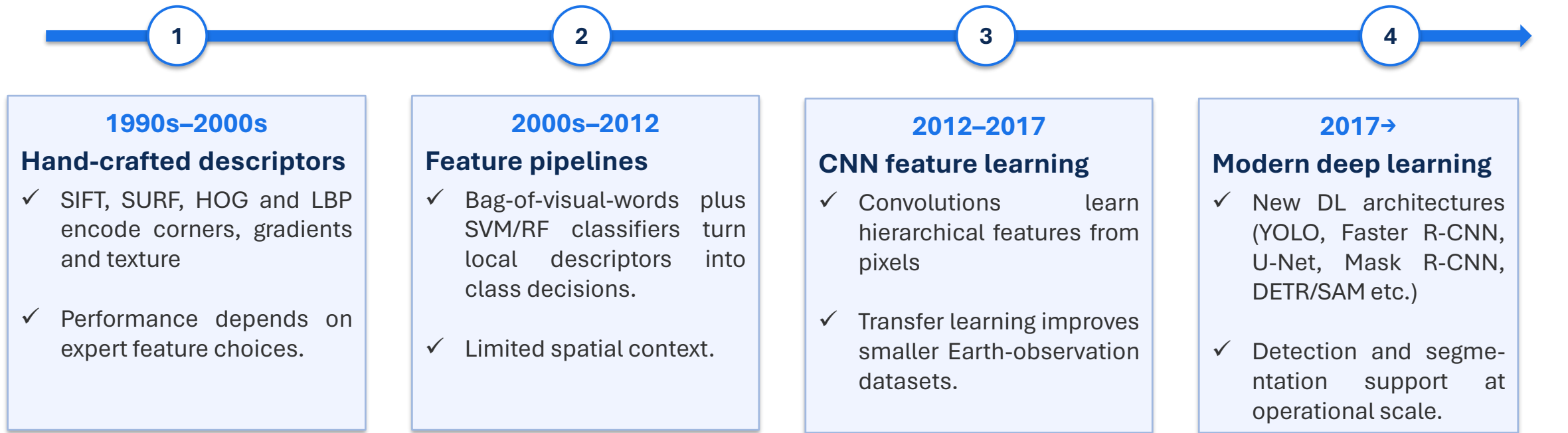
- Viewpoint shift:** Vertical imagery differs sharply from ground-level training data → significant geometric domain shift.
- Rotation:** Objects appear at arbitrary angles → axis-aligned boxes waste area.
- Scale variation:** Targets' sizes range from a few (10-20) to thousands of pixels.
- Spectral variation:** Different aerial and orbital sensors, capturing imagery in different parts of the EM spectrum
- Crowding:** Dense clusters create touching or overlapping instances.
- Annotation cost:** Pixel-level annotation is costly & scarce



Background in Computer Vision for Earth Observation

From hand-crafted descriptors to learned representations

Aerial object recognition evolved from expert-designed texture and shape cues to end-to-end models that learn robust features directly from imagery.

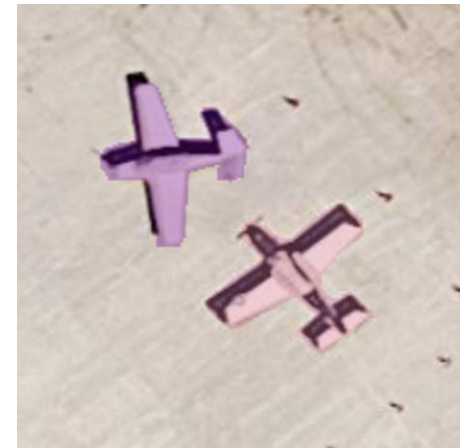
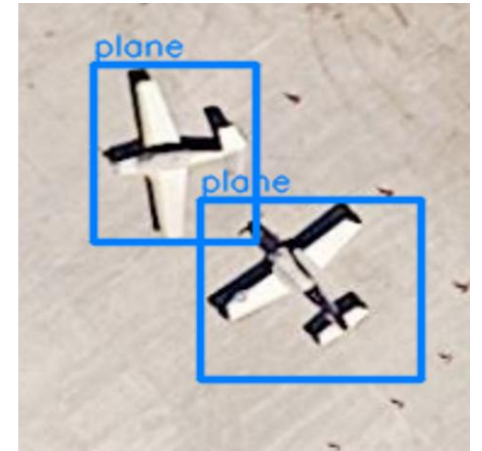
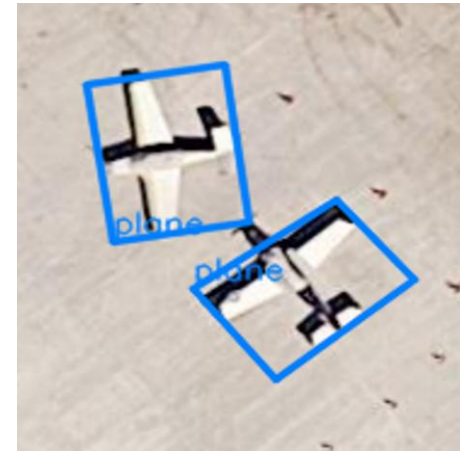


Progressive shift: Manual feature engineering → representation learning → end-to-end detection and segmentation for complex aerial scenes

Background in Computer Vision for Earth Observation

Core CV Tasks

- ✓ **Object Detection:** Localise instances with bounding boxes → Vertical BBox (VBB) or Oriented BBox (OBB).
- ✓ **Semantic Segmentation:** Same mask label to all class objects (no instance distinction).
- ✓ **Instance Segmentation:** Different mask label per individual instance both in the same class and between classes.



Background in Computer Vision for Earth Observation

Detection: From two-stage to one-stage detectors

Distinction: Two-stage detectors propose regions then classify/refine. One-stage detectors predict boxes and classes densely from feature maps.

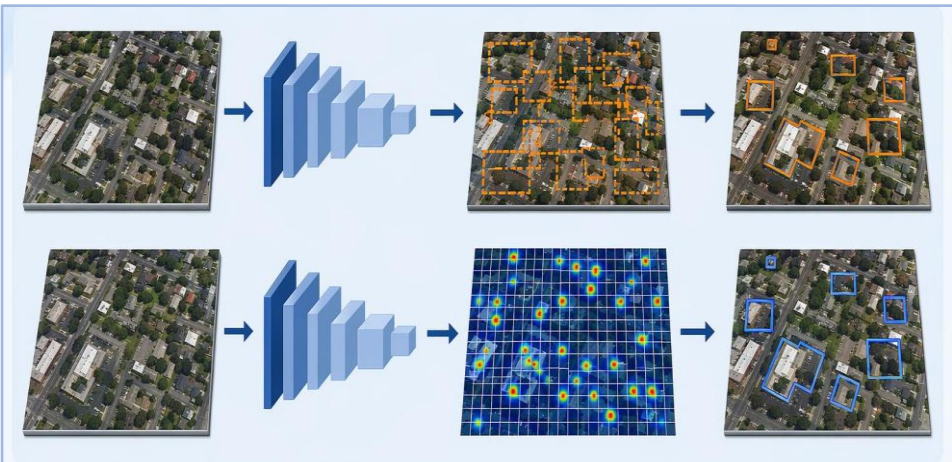
Two-stage detectors

- ✓ Faster R-CNN and Mask R-CNN use region proposals.
- ✓ Strong localization on small objects.
- ✓ Heavier inference pipelines.



One-stage detectors

- ✓ YOLO and RetinaNet predict directly on image derived dense features.
- ✓ Fast inference.
- ✓ Production/operational workflows support.



YOLOv1–v3 (2015–18)

- ✓ Single-stage pioneer
- ✓ Darknet backbone
- ✓ Limited small-object recall



YOLOv4–v7 (2020–22)

- ✓ CSP, PANet, anchor-free (v6)
- ✓ Community OBB extensions
- ✓ 50 - 60% mAP on DOTA



★ YOLOv8 (2023)

- ✓ Native OBB support
- ✓ Ultralytics framework
- ✓ First-class aerial task



YOLO11 (2024)

- ✓ Architecture refinements
- ✓ Higher Precision
- ✓ Improved efficiency



YOLOv12 (2025)

- ✓ Attention-based
- ✓ State-of-the-art VBB on COCO
- ✓ No OBB capability

OBB as a first-class task (YOLOv8+) is a critical enabler for reliable aerial object detection.

Background in Computer Vision for Earth Observation

Detection: Why detection box orientation matters for aerial imagery

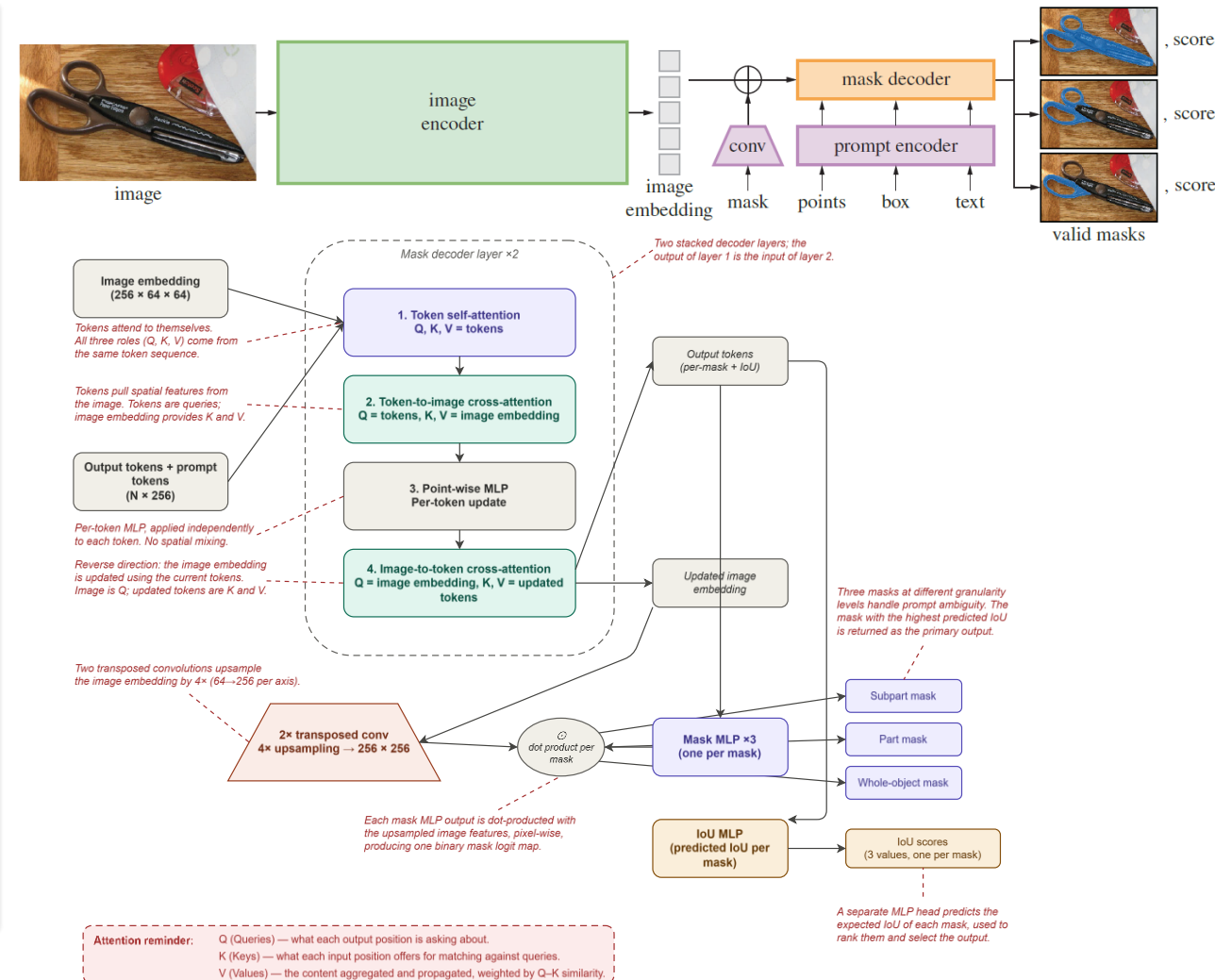
- ✓ VBB overestimate rotated objects by including large background areas → useful pixel coverage can fall below 50% on long, thin targets (e.g. bridges, ships etc.)
- ✓ OBB parameterize rotation directly, tightening the fit → cleaner detection signal.
- ✓ YOLO-OBB and other detectors such as Oriented R-CNN and ReDet are built for this regime.
- ✓ This thesis evaluates VBB and OBB variants separately to quantify their contribution to detection results.



Background in Computer Vision for Earth Observation

Segmentation: The Segment Anything Model (SAM)

- ✓ **Foundational model**
- ✓ **Vision Transformer** encoder (ViT-B / ViT-L / ViT-H: 91M - 630M params) & lightweight prompt encoder & mask decoder.
- ✓ **Trained** on SA-1B: 11M diverse images, >1 billion masks → largest segmentation dataset to date.
- ✓ **Prompt-based**: given a seed point, bounding box or text input → produces a pixel-accurate mask with NO per-dataset training.
- ✓ **Zero-shot** transfer capability makes it an ideal candidate aerial imagery segmentation



Public Aerial Datasets

Detection datasets

General-purpose detection

DOTA v1.0 and DIOR define broad multi-class aerial detection.

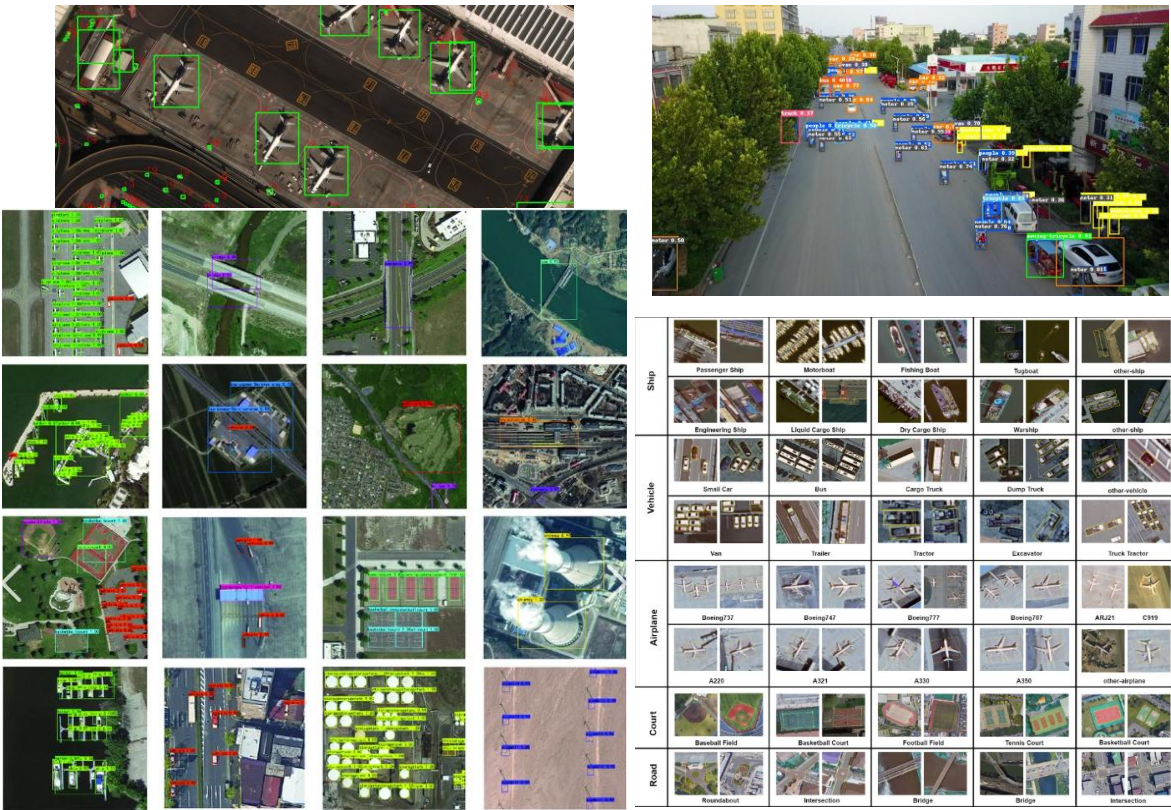
Fine-grained / large-scale

FAIR1M and xView push category depth and instance volume.

Task-specific collections

VisDrone, AI-TOD and NWPU stress viewpoint or tiny-object regimes.

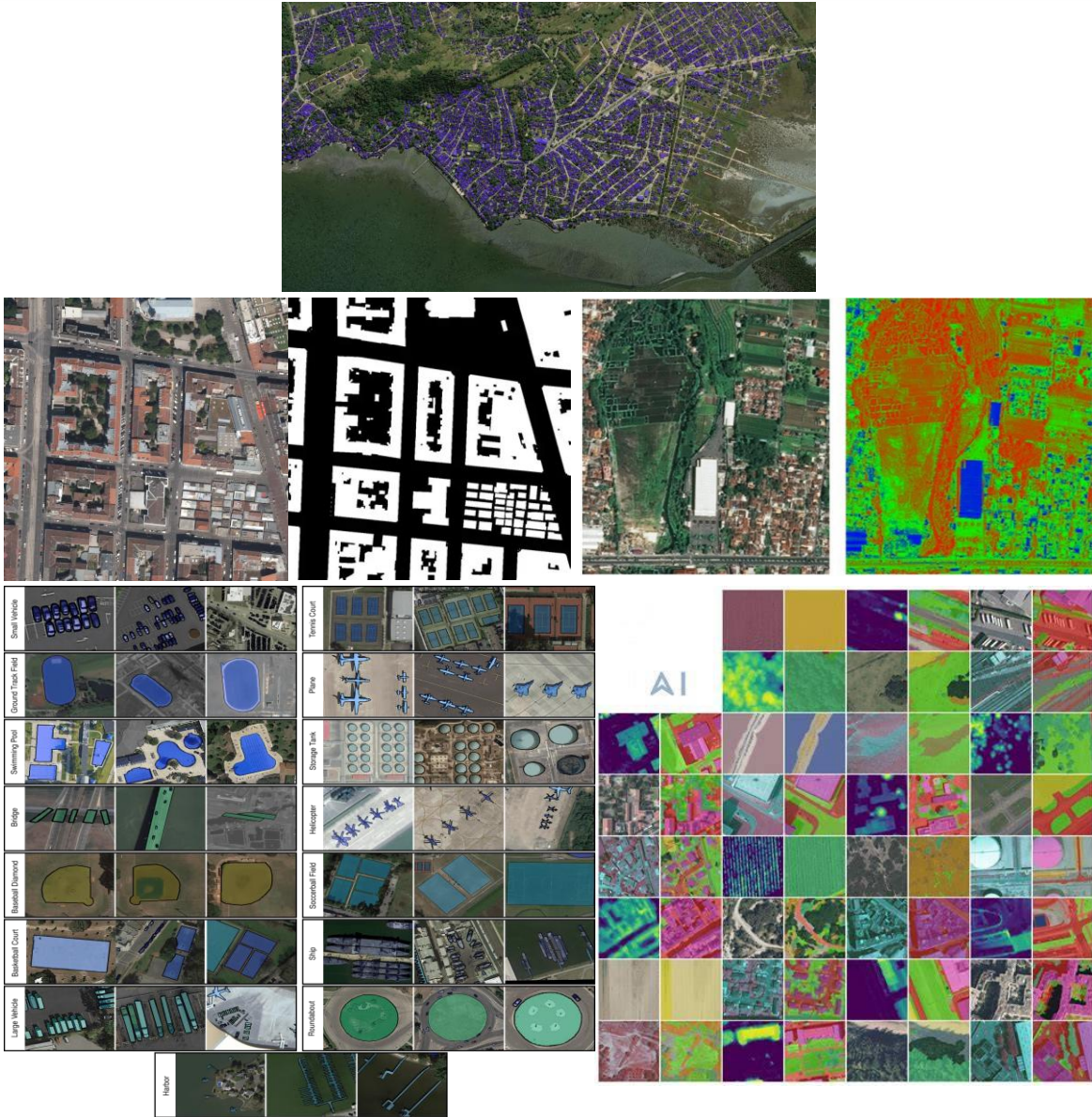
Dataset	Scale	Labels	Research role / caveat
DOTA v1.0	2806 imgs , 188k inst.	15 , OBB	De facto oriented detection standard, multi-scale GSD.
DIOR	23463 imgs , 193k	20 , VBB	Broader compound objects, uniform 800×800 image chips.
FAIR1M	>15k imgs , ~1M	5 / 37 , OBB	Fine-grained aircraft, ship and vehicle recognition.
xView	~1400 km ² , >1M	60 , VBB	WorldView-3 RGB, rich ontology but noisier annotations.
VisDrone	10209 imgs + video	10 , VBB	UAV / oblique traffic-like views, not satellite nadir imagery.
AI-TOD	28036 imgs , 701k	8 , VBB	Tiny-object benchmark, mean object size ≈ 12.8 px.
NWPU VHR-10	800 imgs , 3.7k	10 , VBB	Legacy format benchmark, too small for training from scratch.



Public Aerial Datasets

Segmentation datasets

Dataset	Labels / scale	Role / limitation
iSAID	15 classes , 655k masks	DOTA images with instance masks, but no detection-format labels.
SpaceNet	Buildings / roads / floods	GeoTIFF + georeferencing, narrower task and city coverage.
LoveDA	7 semantic classes	Urban–rural domain adaptation, RGB-only and moderate size.
Inria / ISPRS	Binary or 6-class masks	High-resolution orthophotos, small or single-purpose benchmarks.
FLAIR-1/2	19 land-cover classes	Large-scale French aerial data, FLAIR-2 adds Sentinel-2 time series.



Public Aerial Datasets

DOTA v1.0 EDA

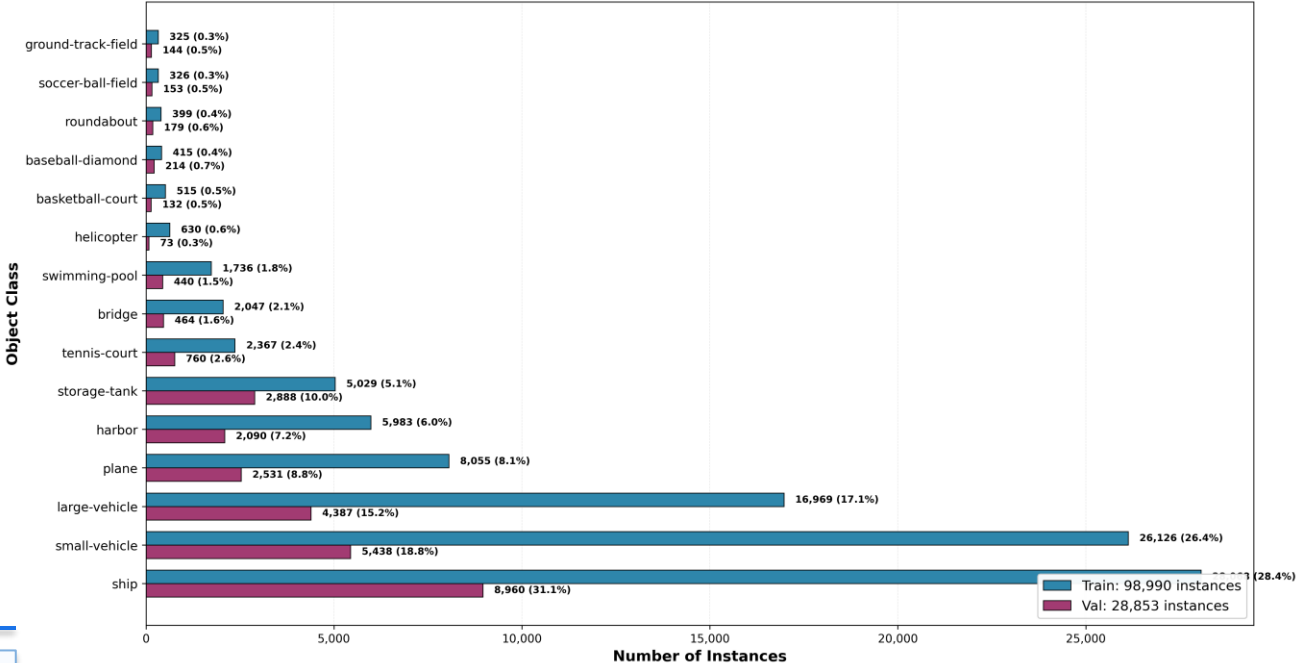
DOTA v1.0 – Object Detection

- ✓ Image sizes: 800×800 to 20000×20000 px
- ✓ 127843 annotated instances , 15 object categories
- ✓ Dual annotation: OBB (4-point polygon) + VBB (axis-aligned)
- ✓ De-facto standard for aerial object detection research
- ✓ Train: 1411 imgs , Val: 458 imgs , Test: 937 imgs (no GT)

15 object classes grouped by target type

Air	plane, helicopter
Maritime	ship, harbor
Vehicles	large vehicle, small vehicle
Urban	bridge, storage tank, roundabout
Sports	baseball diamond, tennis court, basketball court ground-track field, soccer-ball field, swimming pool

DOTA v1 Dataset: Train vs Validation Split Distribution
Total: 127,843 instances (Train: 98,990 | Val: 28,853)



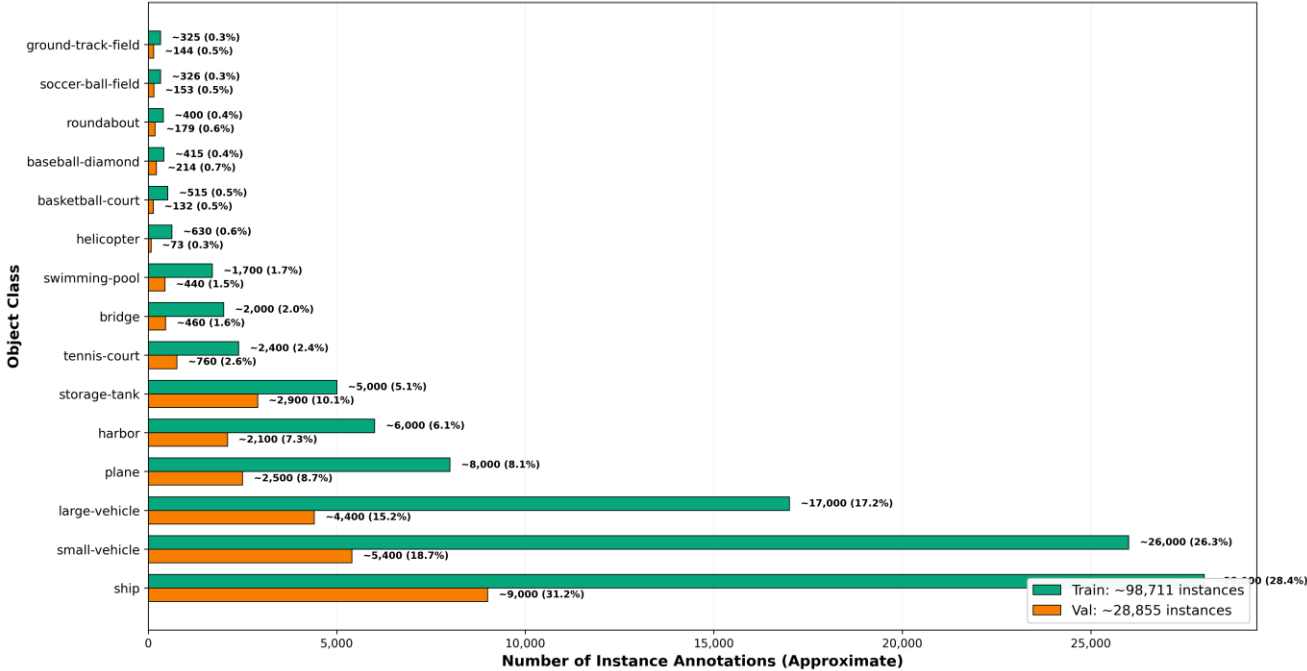
Public Aerial Datasets

iSAID EDA

iSAID – Instance Segmentation

- ✓ Built on top of the same DOTA v1.0 images
- ✓ Semantic & instance segmentation color – coded masks
- ✓ Same 15 object categories
- ✓ Largest publicly available aerial instance segmentation dataset
- ✓ DOTA and iSAID were created independently → annotation inconsistencies → fusion required

iSAID Dataset: Train vs Validation Instance Distribution
Total Instances: ~127,566 (Train: ~98,711 | Val: ~28,855)



Cass masks RGB color coding

1 plane	2 ship	3 storage tank	4 baseball diamond	5 tennis court
6 basketball court	7 ground-track field	8 harbor	9 bridge	10 large vehicle
11 small vehicle	12 helicopter	13 roundabout	14 soccer-ball field	15 swimming pool

background



Research Gaps & Thesis Positioning

Gap 1

No systematic SAM evaluation on DOTA-class aerial objects across all 3 variants and both prompt strategies

Gap 2

No comprehensive YOLO benchmark for modern versions (v8–v12) on DOTA v1.0 under identical conditions

Gap 3

No end-to-end evaluation of a modular YOLO+SAM pipeline on a multi-class aerial benchmark

Gap 4

No unified paired detection + segmentation dataset for aerial imagery compatible with both YOLO and SAM

Gap 5

Limited analysis of prompt strategy effects on aerial segmentation quality

Thesis positioning:

Practical evaluation (not a new architecture). A rigorous investigation of whether state-of-the-art off-the-shelf models can form a viable training-free pipeline for aerial imagery analysis

Research Questions & Thesis Scope

RQ 1

Detection domain transfer

Can COCO-pretrained YOLO models generalize to aerial imagery without domain-specific training?

RQ 2

Segmentation domain transfer

Can SAM's zero-shot segmentation generalise to aerial imagery without fine-tuning?

RQ 3

SAM prompt-strategy effect

How does prompt type (box vs. point) affect SAM segmentation quality for aerial objects?

RQ 4

End-to-end pipeline feasibility

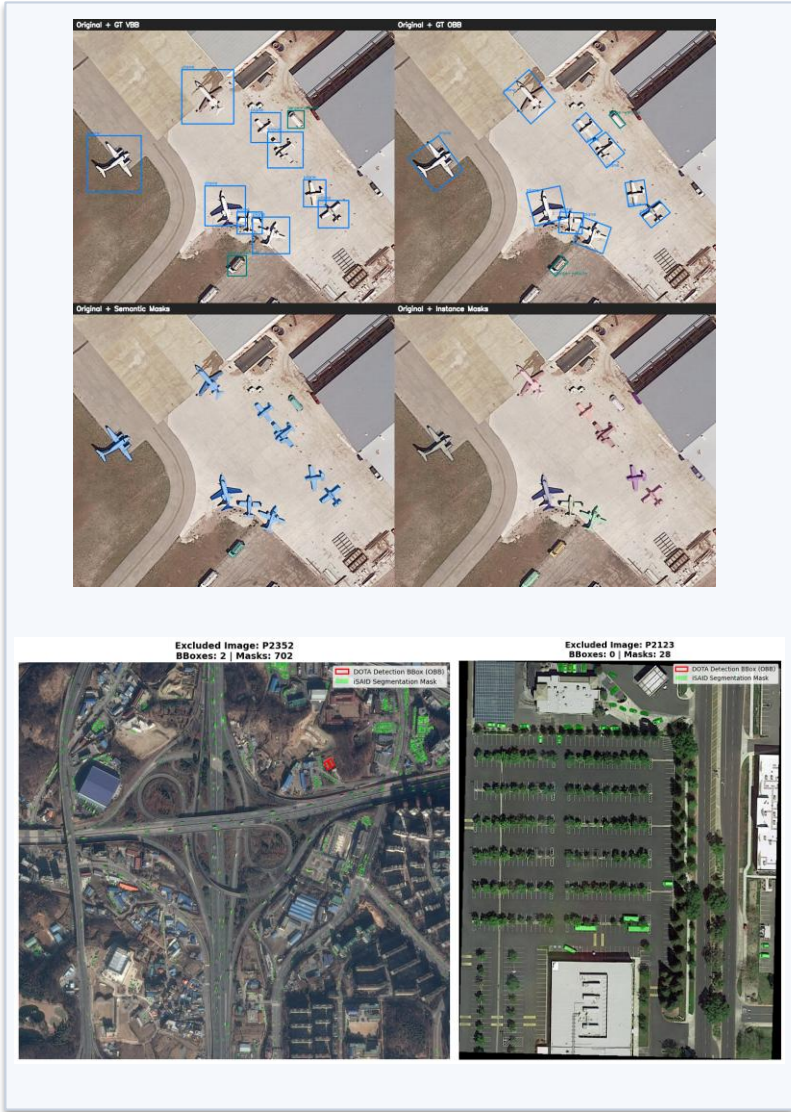
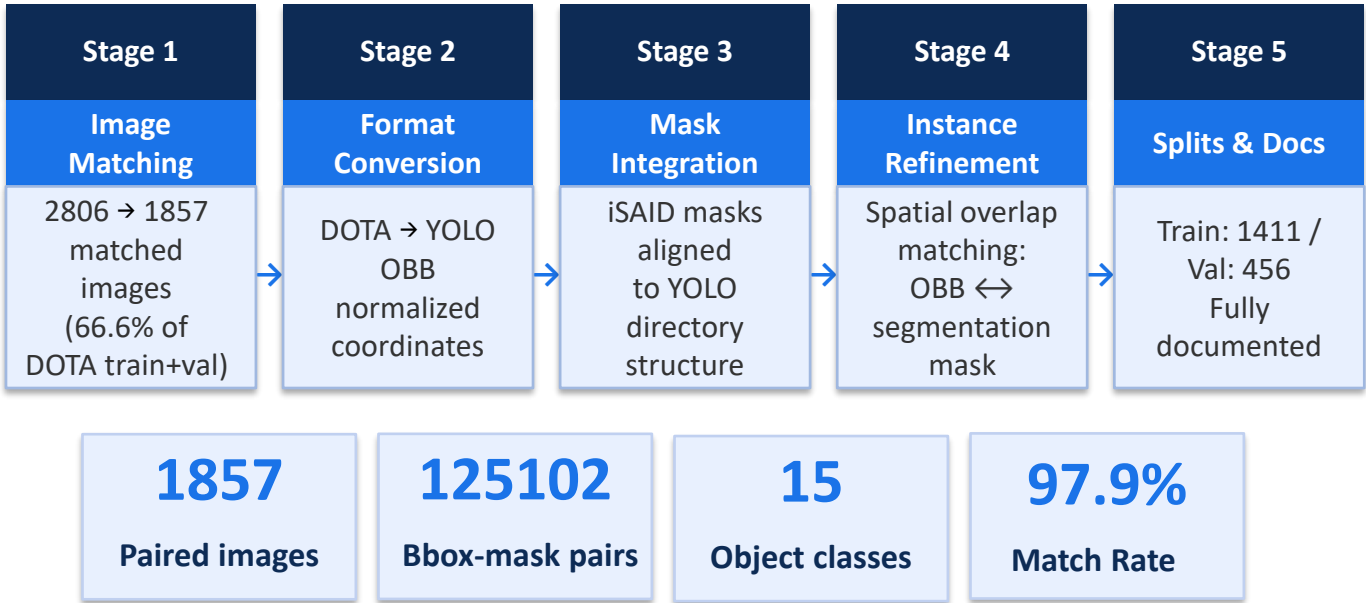
Is a modular YOLO → SAM pipeline viable for operational, real time aerial imagery analysis?

AerialFuseCV – Custom Dataset Creation

Fusing DOTA v1.0 & iSAID into a unified, paired detection-segmentation dataset

Why a new dataset?

✓ 1	Paired labels missing Detection datasets and Segmentation datasets, but no paired data for end-to-end evaluation.	✗ 3	Spectral information discarded Most public benchmarks are 8-bit RGB, losing NIR/SWIR and radiometric depth.
✓ 2	Independent annotation policies Fusion requires instance-level matching, not file concatenation.	✗ 4	Georeferencing absent Outputs remain in pixel space unless externally projected into a CRS.



3 independent experimental sets & 41 total experiments

Exp Set 1 YOLO Baseline

- ✓ 24 COCO-pretrained VBB models (YOLOv8 - v12).
- ✓ 10 DOTA pretrained OBB models (v8, v11).
- ✓ Dataset: DOTA v1.0 validation split.
- ✓ Metrics: Precision, Recall, F1, IoU, DICE, inference time.

GOAL

Isolate YOLO

Domain gap, OBB advantage, quality

Exp Set 2 SAM Baseline

- ✓ 3 SAM variants (ViT-B, ViT-L, ViT-H).
- ✓ 2 prompt strategies (box & point) → 6 configs.
- ✓ Dataset: AerialFuseCV validation split.
Metrics: mask IoU, DICE, inference time.

GOAL

Isolate SAM

Mask quality without detector error

Exp Set 3 Integrated Pipeline

- ✓ Best YOLO → Best SAM pipeline.
- ✓ YOLO predicted boxes used as SAM prompts.
- ✓ Dataset: AerialFuseCV validation split.
Metrics: End-to-end Precision, Recall, F1, IoU, DICE, inference time, Δ IoU.

GOAL

Evaluate pipeline

Error propagation and feasibility

Methodology – Experimental Design

Evaluation framework: Metrics' definitions and interpretation

0  1 best

Exp Set 1 | YOLO Baseline

Classification quality

Precision	Recall	F1 Score
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Correct positives • Detected actual objects • Harmonic mean of P/R
Compute metrics per class

Localization quality

BBox IoU	BBox DICE
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Spatial fit:
>0.5 detected, >0.7 good, >0.9 excellent

IoU = predicted/GT overlap
DICE is less size-sensitive
and better for small objects.

Aggregate means: P, R, F1, IoU and DICE combine into total performance

Exp Set 3 | Integrated Pipeline

Detection & coverage

Micro Precision	Micro Recall	MicroF1
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Prompt correctness • GT coverage (= Coverage) • Harmonic mean of P/R
Count TP, FN, FP, across all classes then compute metrics

Segmentation quality

Mask IoU	Mask DICE
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Count only for what is detected

Mask Δ IoU (SAM Baseline – Integrated Pipeline) to quantify mask quality loss.

Exp Set 2 | SAM Baseline

Mask IoU: Pixel-level overlap of predicted and ground-truth masks
≥0.5 acceptable, ≥0.7 good, ≥0.9 excellent.

Mask DICE: Pixel-level similarity coefficient
emphasizing overlap. More forgiving than IoU.

Mask metrics isolate segmentation quality when prompts/boxes are controlled (perfect).

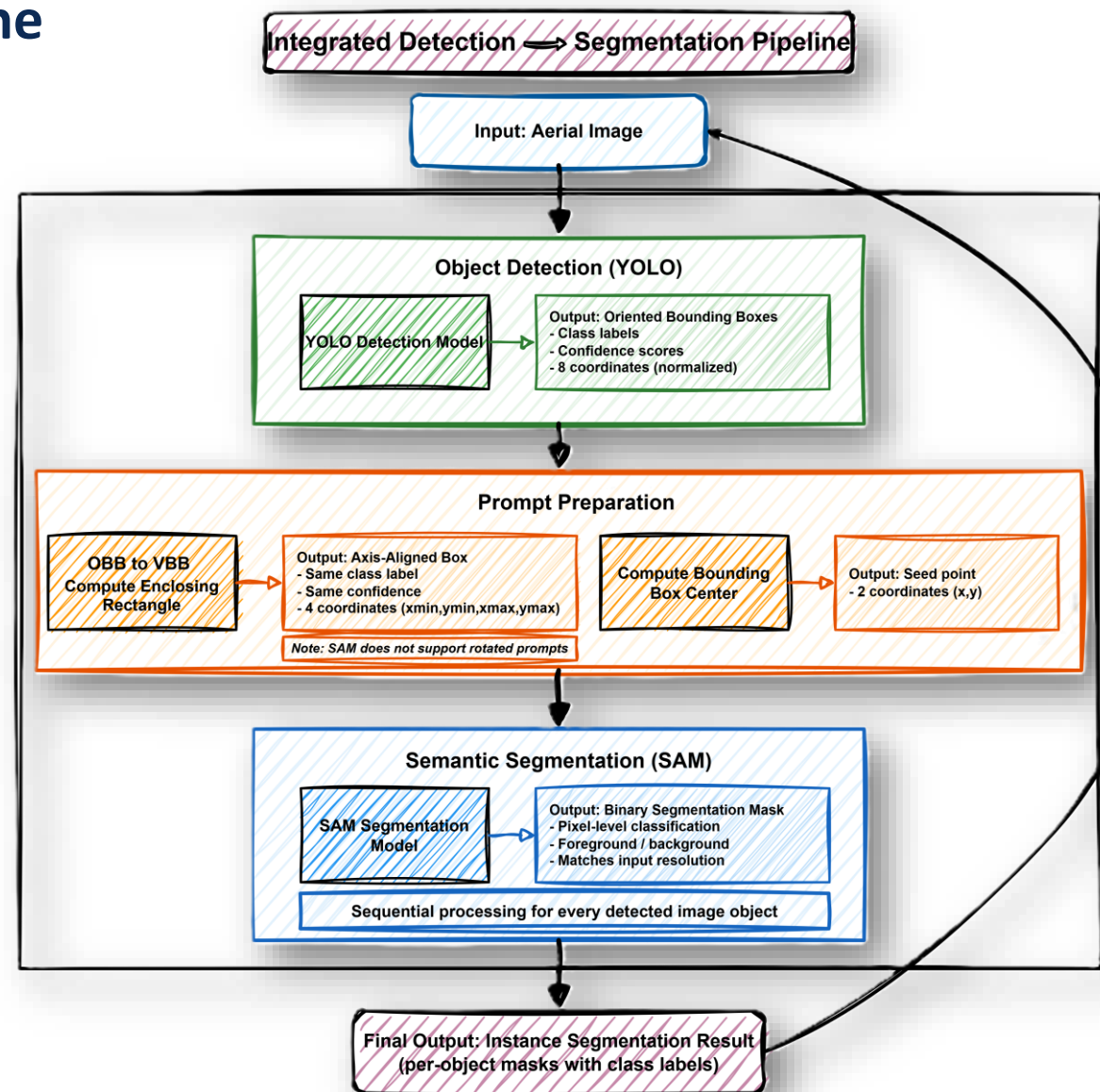
Methodology – Experimental Design

Integrated Detection → Segmentation Pipeline

Pipeline logic

1	Aerial image enters integrated pipeline Single input is passed from detection to segmentation without manual hand-off.
2	YOLO detects candidate objects Returns oriented bounding boxes, class labels, confidence scores and box coordinates.
3	Detections become SAM prompts OBB → VBB conversion and center-point prompts translate detection output into segmentation input.
4	SAM segments each detected object Sequential processing produces segmentation results (masks) per detected object.
5	2 operational modes Evaluation mode: testing and metrics computation Inference mode: operational deployment.

- Detection drives where segmentation is applied.
- Segmentation refines what pixels belong to each object.



Results – YOLO VBB Experiment

24 COCO-pretrained models evaluation

< 0.3%

mean F1 across all models

- ✓ COCO pretraining fails under aerial domain shift
- ✓ Reasons
 - ✗ **Viewpoint mismatch:** Wheels, doors and ground-level shadows learned from COCO are absent from aerial imagery.
 - ✗ **Scale mismatch:** Aerial objects appear at sizes unlike the consumer-photo examples used for pretraining.

Semantic overlap between COCO and DOTA taxonomies in 3/15 classes

COCO car	→	DOTA small-vehicle
COCO boat	→	DOTA ship
COCO truck/bus	→	DOTA large-vehicle

Matched classes isolate domain shift (not class mismatch).

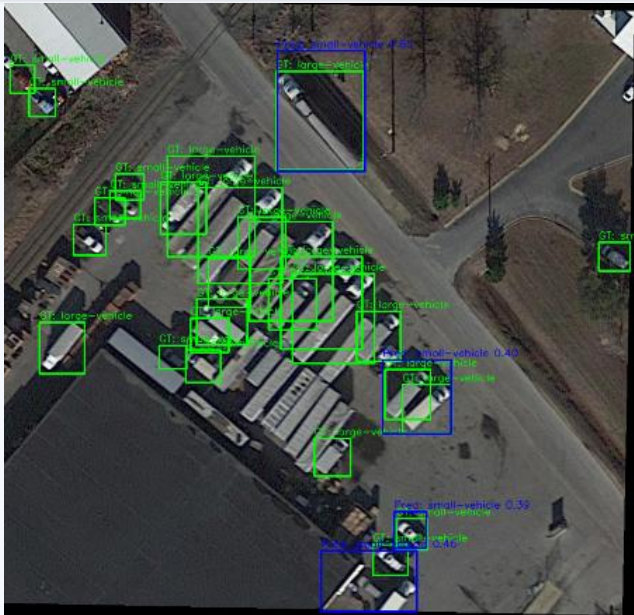
Best model: YOLOv12l-VBB

Class	F1	Recall	Precision	IoU	DICE
ship	0.1%	0.1%	17.6%	0.5%	0.6%
small-vehicle	0.1%	0.0%	28.6%	1.0%	1.2%
large-vehicle	0.0%	0.2%	21.9%	0.6%	0.7%

Findings: Need for domain-specific pretraining or finetuning.

EXAMPLES

— Detection
— Ground truth



Noisy detections persist despite semantic overlap



Results – YOLO OBB Experiment

10 DOTA-pretrained OBB models. Total performance and best model

54.9% - 62.8%

mean F1 range

77.3% - 82.8%

mean Precision range

44.5% - 53.4%

mean Recall range

57.1% - 64.6%

mean Bbox IoU

64.5% - 71.8%

mean Bbox DICE

39.4 - 61.7ms/img

25.4 - 16.2 img/s

Inference Time

RTX 3090 Ti

Best model: YOLOv11x-OBB

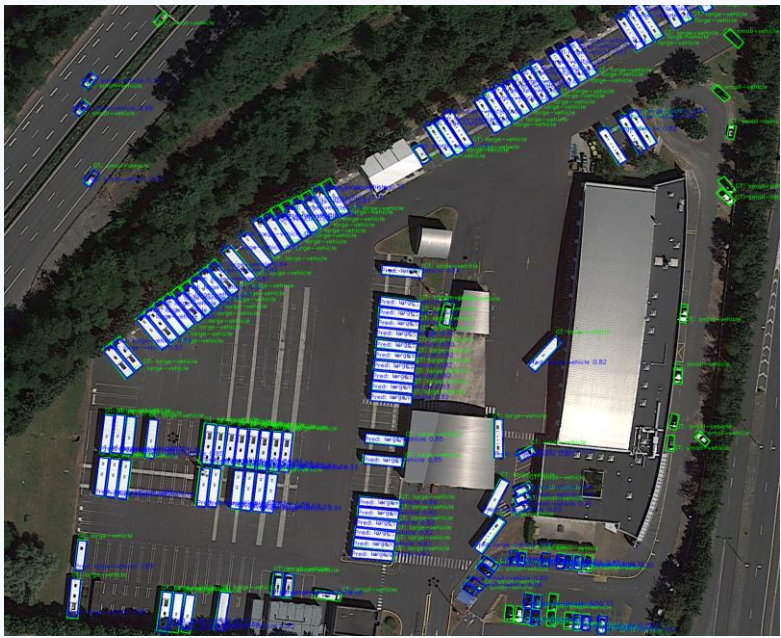
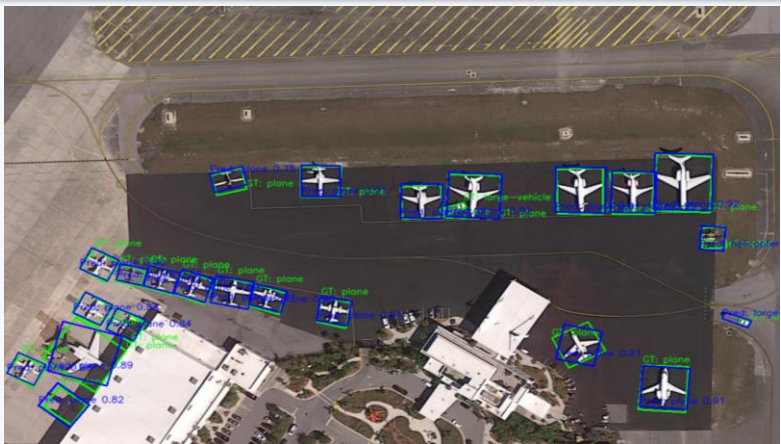
Metric (mean)	Value
F1	62.4%
Recall	52.3%
Precision	82.8%
BBox IoU	64.6%
BBox DICE	71.8%
Time / img	59.8 ms (16.7 img/sec)
Parameters	56.9M

Findings:

- ✓ DOTA-OBB pretraining makes the detector usable.
- ✓ Training regime must be multiscale.
- ✓ YOLOv11x offers the best performance/inference time trade-off.
- ✓ Models are conservative: Less FP (high Precision) but they miss objects (low Recall).
- ✓ DOTA dataset has annotation discrepancies.

EXAMPLES

Detection
Ground truth



Results – YOLO OBB Experiment

YOLOv11x-OBB per-class analysis

High
5 classes

Medium
7 classes

Low
2 classes

Failed
1 class

Findings:

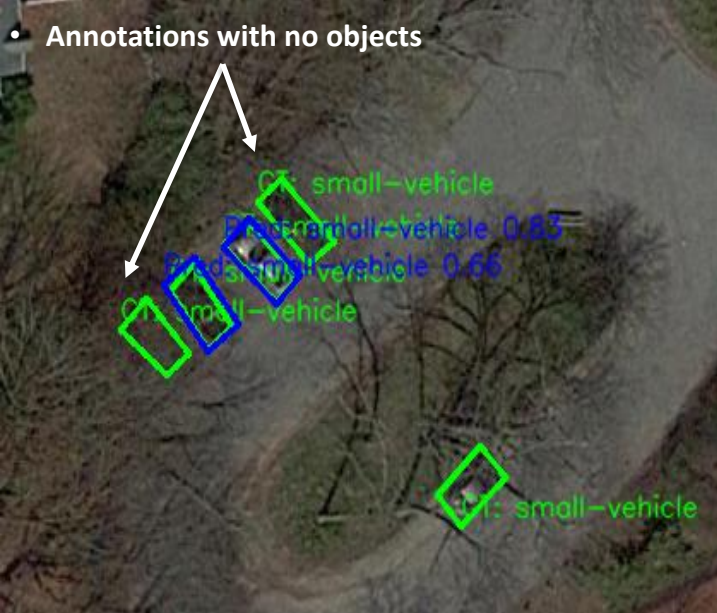
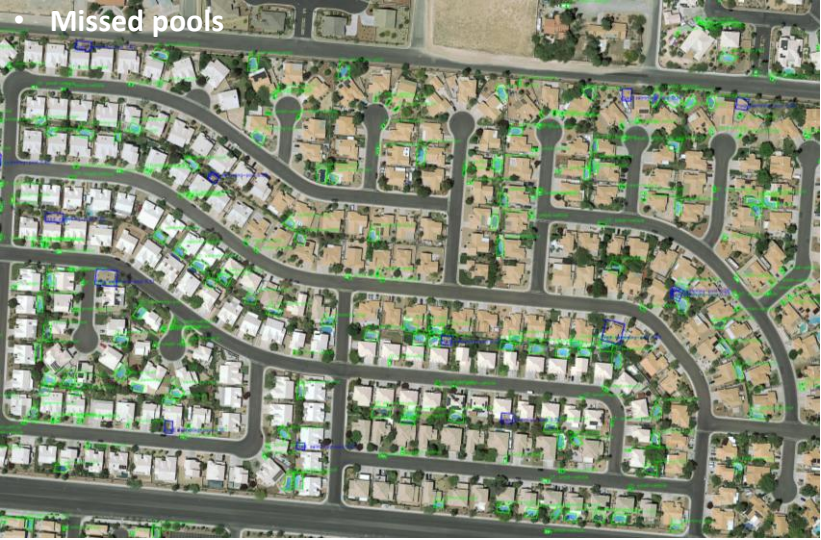
- ✓ Precision high for most classes.
- ✓ Recall is the bottleneck.

Class	F1	Recall	Precision	Det. IoU	Det. DICE
tennis court	93.7%	89.6%	98.1%	92.8%	96.2%
harbor	89.5%	86.8%	92.5%	70.8%	81.8%
large vehicle	85.1%	79.8%	91.1%	81.6%	89.6%
plane	78.4%	66.9%	94.6%	80.8%	89.1%
baseball diamond	75.1%	66.4%	86.6%	73.7%	84.3%
ship	74.2%	62.0%	92.3%	75.6%	85.4%
small vehicle	62.2%	52.9%	75.6%	76.3%	86.2%
basketball court	61.2%	48.5%	83.1%	90.8%	95.1%
ground track	60.3%	48.6%	79.5%	84.6%	91.4%
storage tank	58.1%	41.6%	96.5%	75.4%	85.3%
helicopter	54.9%	38.4%	96.6%	74.4%	85.1%
roundabout	53.4%	37.4%	93.1%	70.9%	81.4%
soccer field	48.8%	38.6%	66.3%	87.2%	92.5%
bridge	40.6%	27.6%	76.6%	63.9%	76.6%
swimming pool	0.4%	0.2%	20.0%	15.1%	26.3%



Results – YOLO OBB Experiment

Failure cases: Model mistakes and annotation errors



Results – SAM Box-Prompted Segmentation Experiment

3 model sizes. Total evaluation

SAM ViT-H
(636M params)

67.7%

79.0%

Mask IoUMask DICE

🕒 943 ms/img or 1.06 img/s

SAM ViT-L
(308M params)

67.5%

78.9%

Mask IoUMask DICE

🕒 800 ms/img or 1.25 img/s

Best model
SAM ViT-B
(91M params)

66.9%

78.5%

Mask IoUMask DICE

🕒 669 ms/img or 1.49 img/s

Class	Mask IoU	Mask DICE
tennis court	83.8%	90.5%
soccer field	81.2%	88.6%
basketball court	75.4%	84.5%
large vehicle	74.9%	84.6%
small vehicle	71.2%	82.6%
swimming pool	69.7%	80.4%
ship	67.5%	79.5%
baseball diamond	67.3%	79.5%
storage tank	64.8%	77.2%
harbor	61.6%	75.2%
roundabout	58.5%	72.1%
ground track	56.7%	70.2%
bridge	54.7%	69.1%
plane	50.0%	65.0%
helicopter	40.5%	55.0%

High
5 classes

Medium
7 classes

Low
2 classes

Failed
1 class

Findings:

✓ SAM generalizes very well from ground to aerial domain. No training or finetuning required → Zero-Shot.

✓ SAM ViT-B offers the best performance/inference time/size trade-off. 7.0× smaller, 0.8% IoU loss, 1.45 × faster.

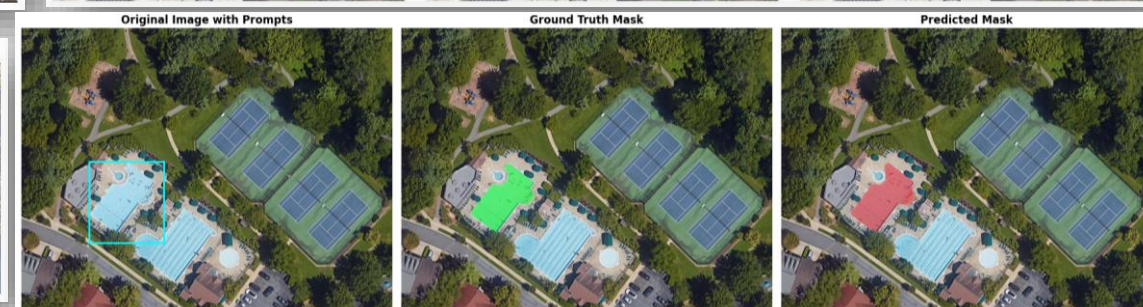
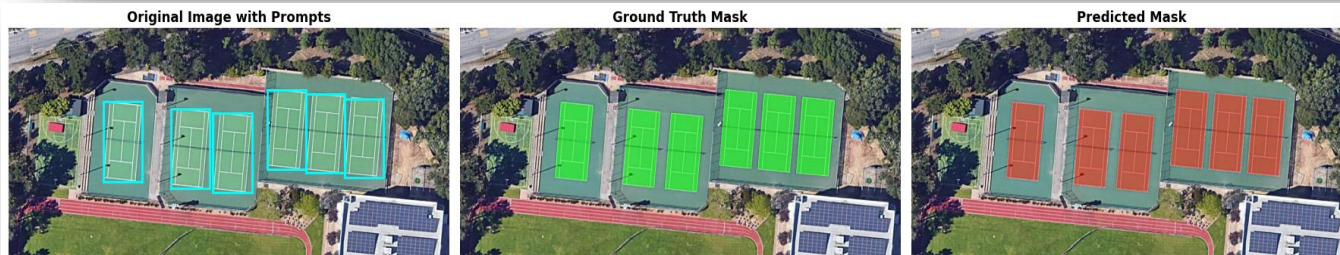
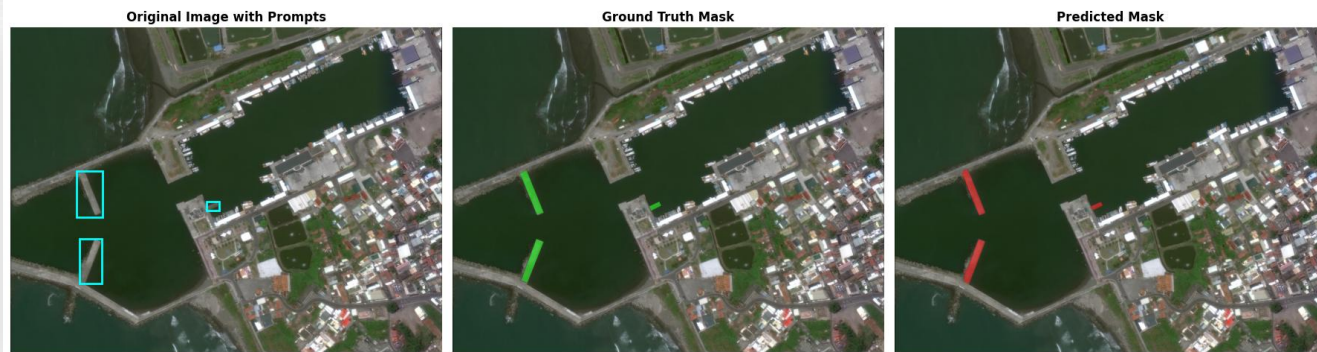
✓ SAM has no batch image processing mechanism.

✓ SAM has no OBB input mechanism → VBBs include much background.

✓ SAM excels on geometrically regular objects (courts, fields, vehicles) and struggles on thin/elongated structures and small objects in small scale images.

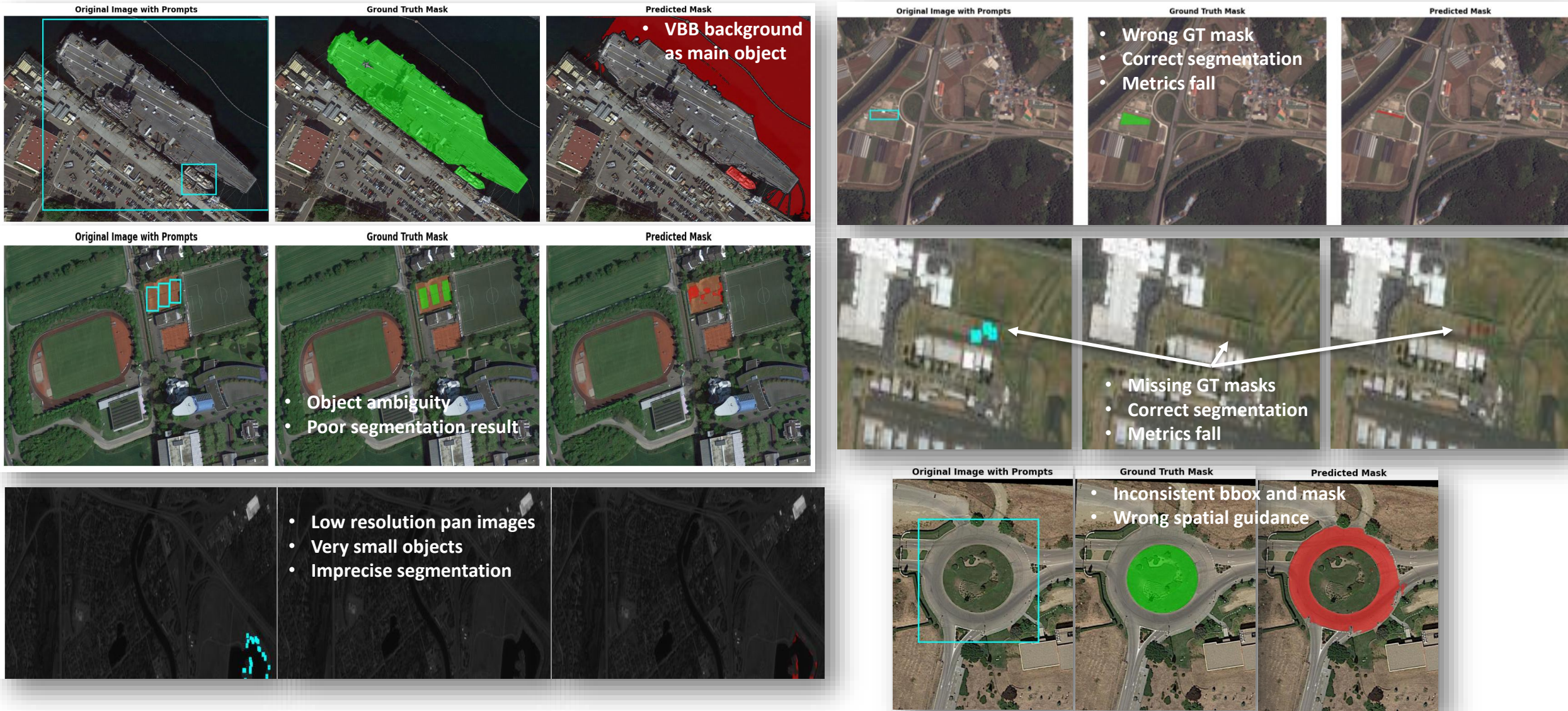
Results – SAM Box-Prompted Segmentation Experiment

Examples



Results – SAM Box-Prompted Segmentation Experiment

Failure cases: Model mistakes, annotation discrepancies and challenging images



Results – SAM Point-Prompted Segmentation Experiment

3 model sizes. Centroid point of bbox as prompt.

SAM ViT-H
(636M params)

47.3%

57.2%

Mask IoU

Mask DICE

🕒 1015 ms/img or 0.98 img/s

SAM ViT-L
(308M params)

47.5%

57.5%

Mask IoU

Mask DICE

🕒 853 ms/img or 1.17 img/s

SAM ViT-B
(91M params)

46.6%

56.8%

Mask IoU

Mask DICE

🕒 724 ms/img or 1.38 img/s

Class	Mask IoU	Mask DICE
tennis court	74.6%	83.5%
large vehicle	65.2%	75.4%
small vehicle	62.7%	74.5%
swimming pool	59.3%	69.5%
ship	47.9%	59.4%
soccer field	47.8%	57.3%
baseball diamond	47.2%	58.5%
plane	44.0%	55.7%
basketball court	38.4%	46.6%
roundabout	36.7%	43.9%
ground track	35.2%	44.4%
harbor	34.4%	45.4%
storage tank	32.3%	43.3%
helicopter	29.4%	37.7%
bridge	12.8%	18.3%

High
0 classes

Medium
4 classes

Low
4 classes

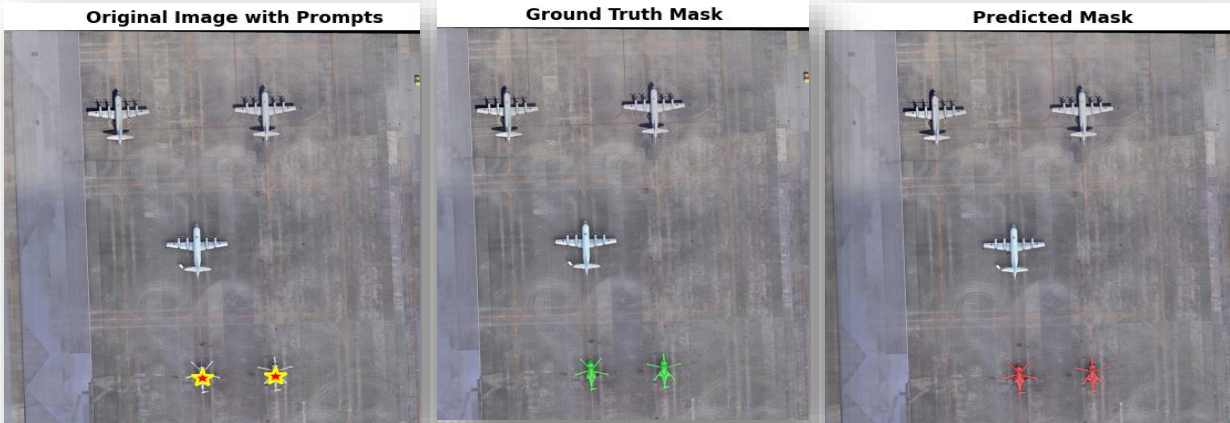
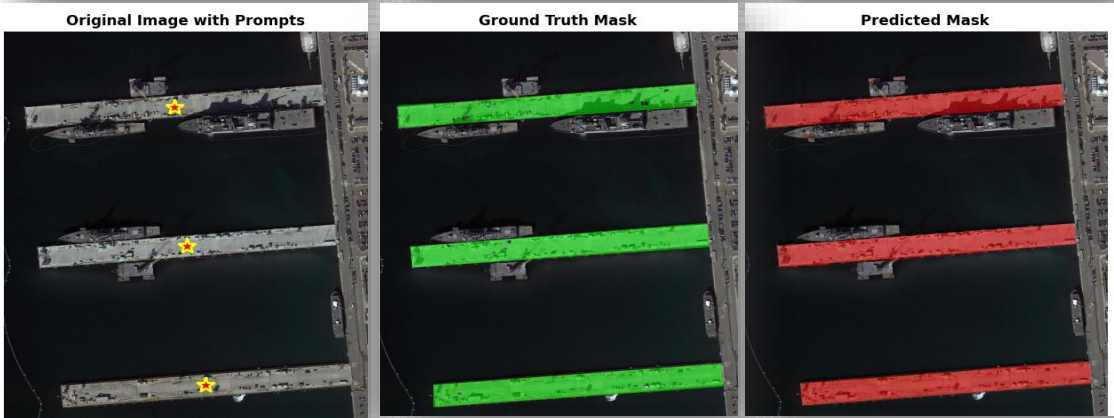
Failed
7 class

Findings:

- ✓ Model size is secondary: 91M→630M adds only +0.8% IoU.
- ✓ Point prompts lose spatial context. Failures cluster on irregular, elongated, or rotated objects.
- ✓ Point placement is crucial, but difficult to automate. Can direct to object sub-part or in background → centroid not always the best choice (object shape, rotation etc.)
- ✓ OBB→VBB conversion can add background noise, limiting current SAM point-prompt performance.

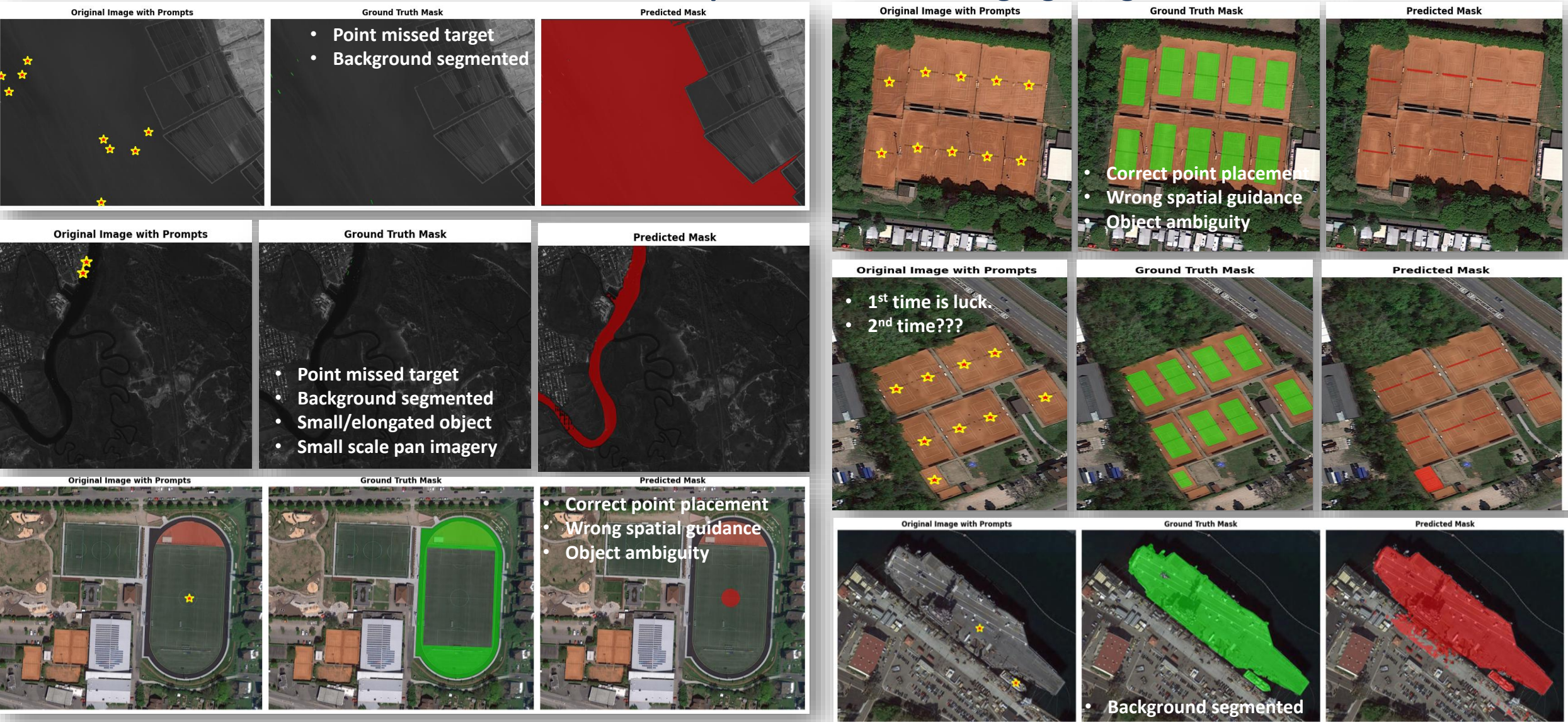
Results – SAM Point-Prompted Segmentation Experiment

Examples



Results – SAM Box-Prompted Segmentation Experiment

Failure cases: Model mistakes, annotation discrepancies and challenging images



Results – Integrated Detection→Segmentation Pipeline

YOLOv11x-OBB + SAM-ViT-B-BOX: Total evaluation

81.5%

Detection micro Precision
(end-to-end)

62.1%

Detection micro Recall
(Coverage)

70.5%

Detection microF1
(end-to-end)

67.9%

Segm. IoU
(matched pairs)

79.5%

Segm. DICE
(matched pairs)

786.7 ms/img
1.27 img/s
Inference Time
(end-to-end)
RTX 3090 Ti



Findings:

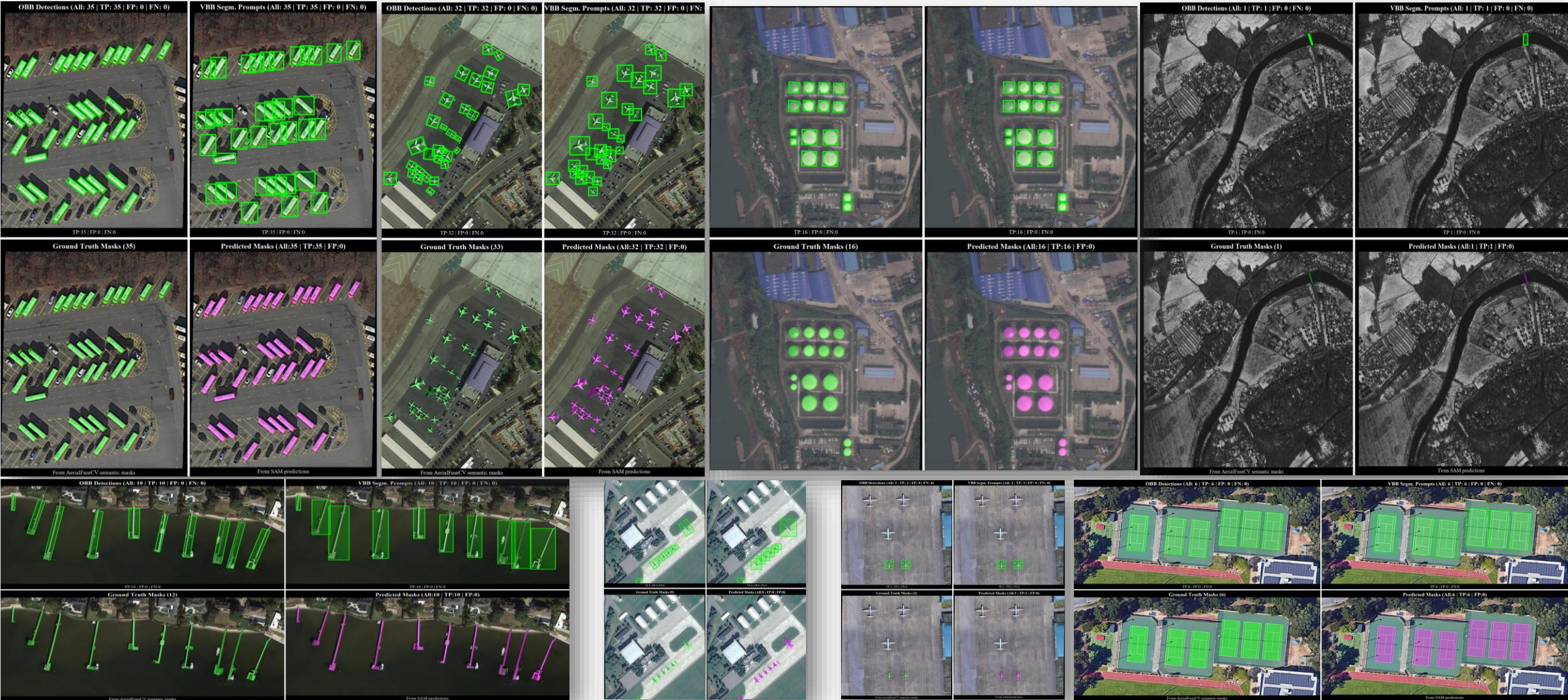
- ✓ End-to-end pipeline reaches 70.5% F1 detection, 67.9% IoU / 79.5% DICE segmentation on matched pairs.
- ✓ 62.1% Recall means 37.9% of GT instances never receive a prompt → SAM cannot recover what YOLO doesn't propose.
- ✓ Asymmetric P/R (81.5% vs 62.1%) reveals a conservative detector → when it fires, it's usually right, but it fires too rarely.
- ✓ Segmentation quality equivalent to SAM baseline experiment.
- ✓ Compute budget dominated by SAM (77.5%)

Per-Class Analysis

Class	Detection Stage			Segmentation Stage	
	Precis.	Recall	F1	IoU	DICE
soccer-ball-field	65.2	41.1	50.4	87.1	92.2
tennis-court	97.7	90.4	93.9	85.6	91.5
basketball-court	70.1	47.0	56.2	82.2	89.1
large-vehicle	80.6	82.9	81.7	79.4	87.7
storage-tank	87.8	46.6	60.8	71.8	82.1
small-vehicle	69.2	58.5	63.4	70.6	81.6
baseball-diamond	86.6	66.4	75.1	67.6	79.7
ship	83.7	63.0	71.9	63.2	76.2
harbor	83.6	62.2	71.3	61.1	74.6
roundabout	88.9	35.4	50.6	60.5	73.7
ground-track-field	78.4	48.9	60.3	58.5	72.0
bridge	74.3	27.6	40.3	56.2	70.5
plane	91.1	63.0	74.5	54.1	69.4
helicopter	72.4	31.3	43.8	41.0	55.9
swimming-pool	0.0	0.0	0.0	0.0	0.0

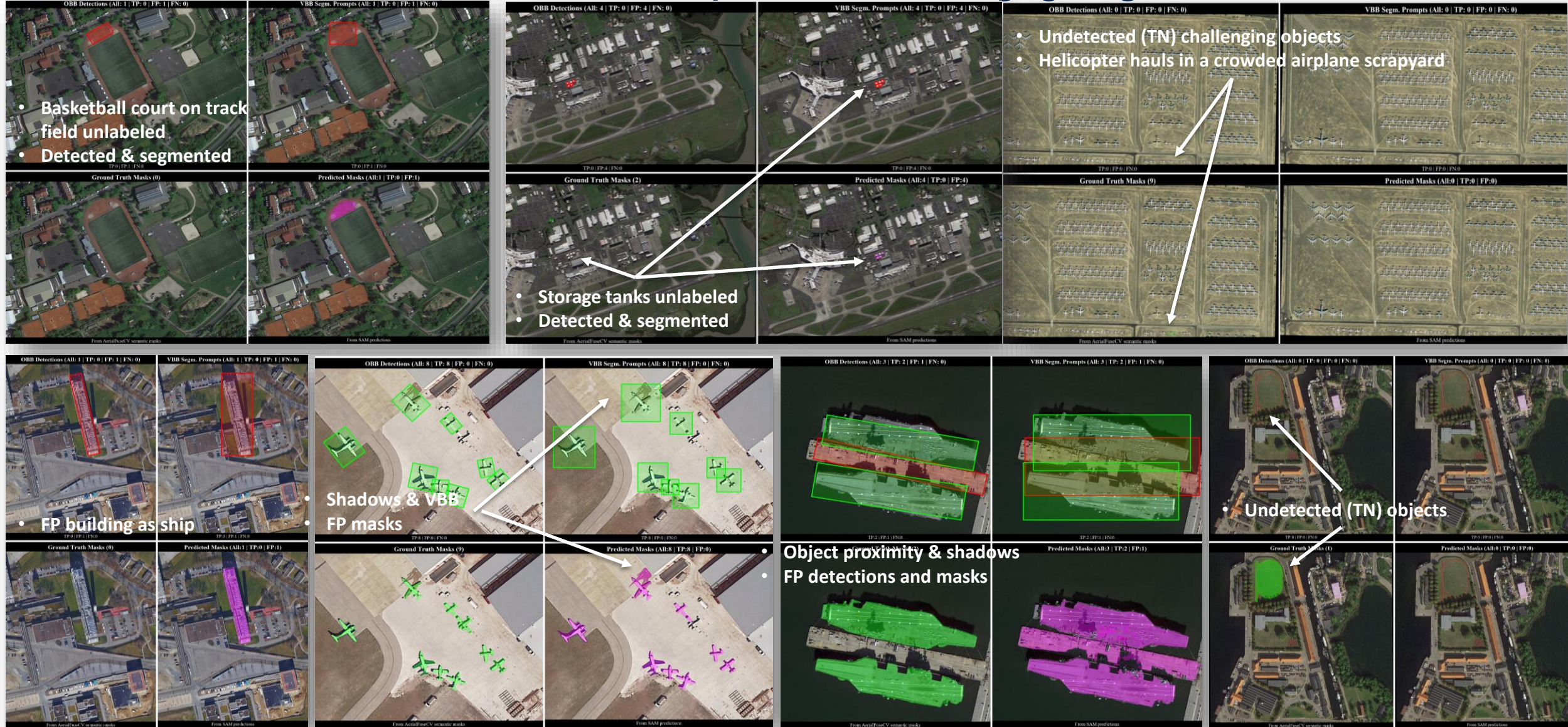
Results – Integrated Detection→Segmentation Pipeline

Examples



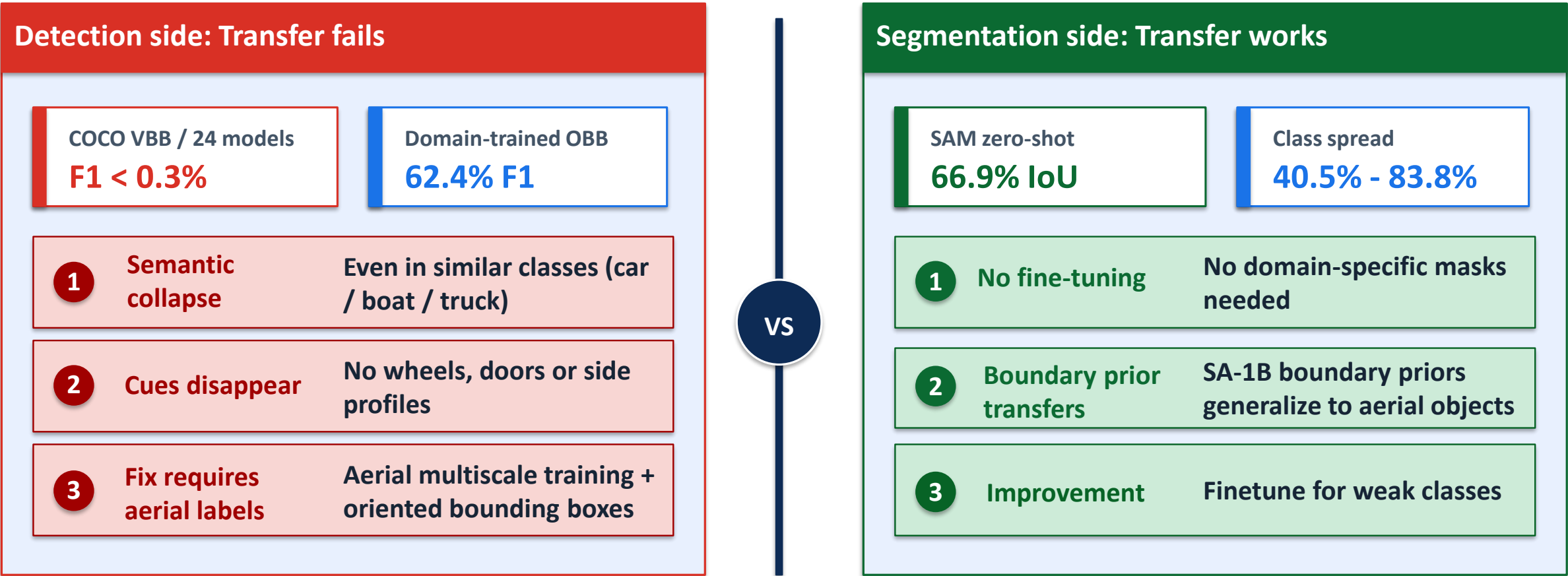
Results – Integrated Detection→Segmentation Pipeline

Failure cases: Model mistakes, annotation discrepancies and challenging images



Discussion – Domain Transfer: Detection vs Segmentation

Sharp asymmetry: Detection breaks but segmentation transfers through Ground → Aerial domain shift



Key conclusion: Detection is domain-dependent. Segmentation is domain-agnostic with SAM.

Discussion – Detection in Aerial Domain

Coverage, not localization, limits the detector

VBB transfer baseline

F1 <0.3%

Best OBB detector

62.4% F1

Detector balance

82.8%P / 52.3%R

Scale gain

7.9%

1. Spatial localization matters

- ✓ VBB boxes are wrong for rotated objects.
- ✓ VBB can enclose ~40–60% background pixels.
- ✓ Detection in crowded scenes degrades.

2. Recall is the main struggle

- ✓ YOLO models are conservative (low Recall).
- ✓ Missed objects are unrecoverable.
- ✓ Detector optimization should favor recall, not precision.

3. Non uniform detector behavior

- ✓ Performance is class dependent and bimodal.
- ✓ 5 classes are deploy-ready (F1 $\geq 75\%$).
- ✓ 1 class is catastrophic (F1 0.4%, Recall 0.2%).

4. Localization quality is decoupled from detection coverage

- ✓ 14 of 15 classes have BBox IoU $\geq 70\%$ when the detector fires
- ✓ Even low-Recall classes produce tight boxes

Key conclusion: Per-class confidence thresholds, recall-oriented & multiscale fine-tuning, and OBB-native annotations are the highest-leverage interventions.

Discussion – Segmentation in Aerial Domain

SAM is zero-shot, but prompt quality and geometry set the limitations

1	Model scale is not important	91M vs 636M → 7× smaller, 1.4× faster, only ~0.8% IoU loss in box-prompt regime.
2	Geometry determines success	Strong: compact regular classes >75%. Weak: Elongated / irregular objects <50%
3	OBB → VBB adds boundary error	SAM box API accepts axis-aligned boxes. Converted OBBs add ~15 - 40% background pixels.
4	Zero-cost upper bound	SAM baseline reaches 66.9% IoU, with no training. Other segmenters (e.g. Mask R-CNN) can do better (70-75% ?), but with extreme cost (domain specific training).
5	Native OBB prompting and batch processing	Would close part of the gap. Requires architectural intervention.
6	Prompt type dominates mask quality & model scale	Spatial context beats scale. Switching box→point costs ~20% IoU no matter the model size.

	High score≥75 gap<10	Medium score 50–74 gap 10–19	Low score 40–49 gap 20–29	Failed score <40 gap≥30		
	Box IoU	Point IoU	IoU Gap	Box DICE	Point DICE	DICE Gap
bridge	54.7%	11.1%	43.6%	69.1%	16.6%	52.5%
basketball court	75.4%	38.2%	37.1%	84.5%	46.8%	37.7%
storage tank	64.8%	30.8%	34.0%	77.2%	41.3%	35.9%
soccer field	81.2%	52.7%	28.4%	88.6%	62.8%	25.8%
roundabout	58.5%	32.2%	26.3%	72.1%	38.6%	33.5%
harbor	61.6%	35.9%	25.7%	75.2%	48.1%	27.0%
ground track	56.7%	34.5%	22.3%	70.2%	44.3%	25.9%
baseball diamond	67.3%	45.7%	21.6%	79.5%	57.0%	22.5%
ship	67.5%	47.1%	20.3%	79.5%	58.4%	21.1%
helicopter	40.5%	27.9%	12.6%	55.0%	36.4%	18.5%
large vehicle	74.9%	63.1%	11.8%	84.6%	73.7%	10.9%
small vehicle	71.2%	59.7%	11.4%	82.6%	72.1%	10.5%
swimming pool	69.7%	60.3%	9.4%	80.4%	70.3%	10.1%
tennis court	83.8%	75.1%	8.7%	90.5%	83.5%	7.0%
plane	50.0%	41.6%	8.4%	65.0%	53.6%	11.4%

Discussion – Integrated Detection→Segmentation Operational Feasibility

Recall bottleneck

62.1%

Detection coverage

- ✓ **37.9%** of ground-truth objects receive NO segmentation prompt.
- ✓ Pipeline IoU = 67.9% BUT missed instances are invisible.
- ✓ +1.0% ΔIoU vs SAM baseline is a selection effect, not a real gain.
- ✓ 8 / 15 classes have Recall <50% (majority missed in most categories).

Real-time suitability

786.7 ms/image or 1.3 images/s

Processing time breakdown

Overhead: 0.7%

YOLO
21.8%

SAM Segmentation: 77.5%

SAM is the bottleneck: **92%** of total compute time.

Deployment scenarios

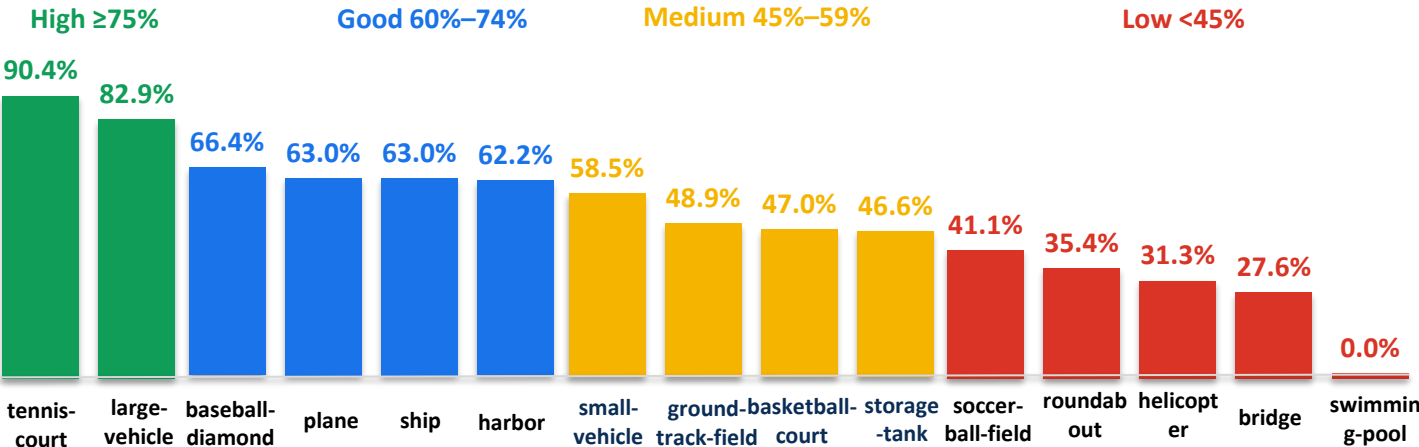
✓ **Suitable** Post-flight UAV analysis. Batch, offline workflow.

✓ **Suitable** Satellite image processing. ~6 min for 456 images.

✓ **Suitable** Offline geospatial mapping. Training-free deployment.

✗ **Not suitable** Real-time UAV (10–30 fps). 10–25× too slow.

Recall distribution across 15 classes



Key strengths for deployment

- ✓ **Modular:** Swap YOLO/SAM independently with models.
- ✓ **No fine-tuning:** Eliminates dataset prep, GPU training time, and labeling cost.
- ✓ **Deployable classes:** 6 classes ready for productivity.
- ✓ **New annotations:** Speed up segmentation dataset augmentation in new classes.

Limitations & Assumptions

Dataset scope

All experiments on DOTA v1.0 / iSAID only. Results on xView, VisDrone, or proprietary UAV collections may differ as sensor distribution not generalized.

OBB → VBB conversion

SAM does not accept rotated box prompts. VBBs include background for elongated and rotated objects (ships, bridges) → degraded spatial guidance for segmenter.

Matching threshold

IoU $\geq$ 0.1 threshold used. Stricter thresholds (0.3, 0.5) would reduce Recall and potentially change per-class rankings. A sensitivity analysis (Recall – Precision curve) is not included.

Zero-shot only

No domain-specific fine-tuning performed. Results represent out-of-the-box capability. Domain fine-tuned models would likely outperform.

Annotation inconsistencies

AerialFuseCV inherits DOTA/iSAID annotation noise (unannotated objects counted as FP, masks extending beyond true boundary, loose GT detection boxes etc.). Impact cannot be precisely quantified.

No supervised segmentation baseline

No comparison to Mask R-CNN / YOLACT / SOLOv2. Cannot quantify annotation-cost trade-off vs fully supervised instance segmentation.

Future Research Directions

High Impact (Short-term)

- ✓ Fine-tune YOLO on DOTA with Recall-optimised loss → close the 37.9% coverage gap.
- ✓ Evaluate SAM2 & EfficientSAM for speed/quality trade-off.
- ✓ Hybrid prompting: box for large objects, point for compact classes (mechanism already integrated).
- ✓ Sensitivity analysis: IoU thresholds 0.1 / 0.3 / 0.5 (Recall-Precision curve).

Medium Impact (Mid-term)

- ✓ Native OBB prompts in SAM to eliminate background noise from VBB conversion.
- ✓ Confidence-based segmentation post or pre filtering to reduce 18.5% FP overhead ($3697 \text{ FPs} \times 610 \text{ ms} = \sim 37$ minutes of wasted SAM inference).
- ✓ Compare vs supervised segmentation baselines (Mask R-CNN, YOLACT, Mask2Former).

Broader Scope (Long-term)

- ✓ Multi-dataset evaluation like xView, VisDrone, custom UAV captures.
- ✓ Edge hardware benchmarks like NVIDIA Jetson, mobile GPUs, quantised models.
- ✓ Temporal extensions like aerial video, object tracking, change detection.
- ✓ Multi-sensor evaluation with SAR, multispectral, hyperspectral imagery.

Contributions of This Thesis

C1

34-Model YOLO Benchmark

A comprehensive and systematic evaluation of YOLOv8 through YOLOv12 in both VBB and OBB configurations on DOTA v1.0 under identical conditions.

C2

6-Config SAM Benchmark

Evaluation of all 3 SAM variants with 2 prompt strategies on 15 aerial object classes, systematically quantifying the prompt quality vs. model capacity trade-off. No previous work on this (no combined dataset).

C3

End-to-End Pipeline

Complete, reproducible and modular Detection→Segmentation pipeline with dual-metric evaluation that separates detection and segmentation performance.

Can contribute to costly segmentation dataset new annotations.

C4

AerialFuseCV Dataset

Fused evaluation dataset from DOTA v1.0 + iSAID providing paired detection and segmentation annotations, enabling integrated pipeline analysis.

Conclusions – Answering the Research Questions

RQ1: Detection domain transfer

VBB COCO-pretrained models → aerial domain: $F1 < 0.3\%$ and NOT viable. OBB DOTA-pretrained: 54.9–62.4% F1. Domain-specific training + OBB representation are hard requirements.

RQ2: Segmentation domain transfer

SAM achieves 66.9% IoU zero-shot. Segmentation is domain-agnostic. Boundary delineation generalizes to aerial views without any fine-tuning.

RQ3: Prompt strategy effect

Box prompts: ~67% IoU. Point prompts: ~47% IoU. Prompt quality (20% gap) dominates over model size (0.8% gap). Box prompts are overall better.

RQ4: Pipeline feasibility

Feasible for batch-mode workflows (1.3 img/s). Not real-time. Detection Recall (62.1% coverage) is the limiting factor, not segmentation quality.

Summary - The big picture in five takeaways

1

Domain gap is real and detection-specific

YOLO needs domain-specific OBB pretraining. SAM does not. Segmentation generalizes, detection does not.

2

Prompt quality matters more than model size

A 20% IoU gain from box vs point prompts more important than any model scaling benefit. Invest in better detection.

3

Detection bottleneck limits the system

62.1% coverage means 4 in 10 objects are invisible. Improving Recall is the highest-leverage improvement.

4

AerialFuseCV enables unified pipeline evaluation

DOTA + iSAID fusion creates the missing infrastructure for paired detection → segmentation benchmarking.

5

Training-free pipeline is operationally viable

1.3 img/s is practical for batch workflows. Zero annotation cost for segmentation is a real advantage.



Thank You

Questions & Discussion



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